SYRIATEL CUSTOMER CHURN

Predicting and Preventing Customer Loss

Data Science Phase 3 Project

March 2026

OVERVIEW

- SyriaTel is facing increasing customer churn, leading to significant revenue loss and reduced customer retention. Identifying customers who are likely to leave has become a critical challenge, making it difficult for the company to take timely and effective retention actions.

PROBLEM STATEMENT

Problem

- Lost revenue from cancelled subscriptions
- Acquiring new customers costs 5-25× more
- Market share loss to competitors

Opportunity

- Identify at-risk customers early
- Enable proactive retention efforts
- Understand what drives churn
- Allocate resources efficiently

TOOLS & TECHNOLOGIES

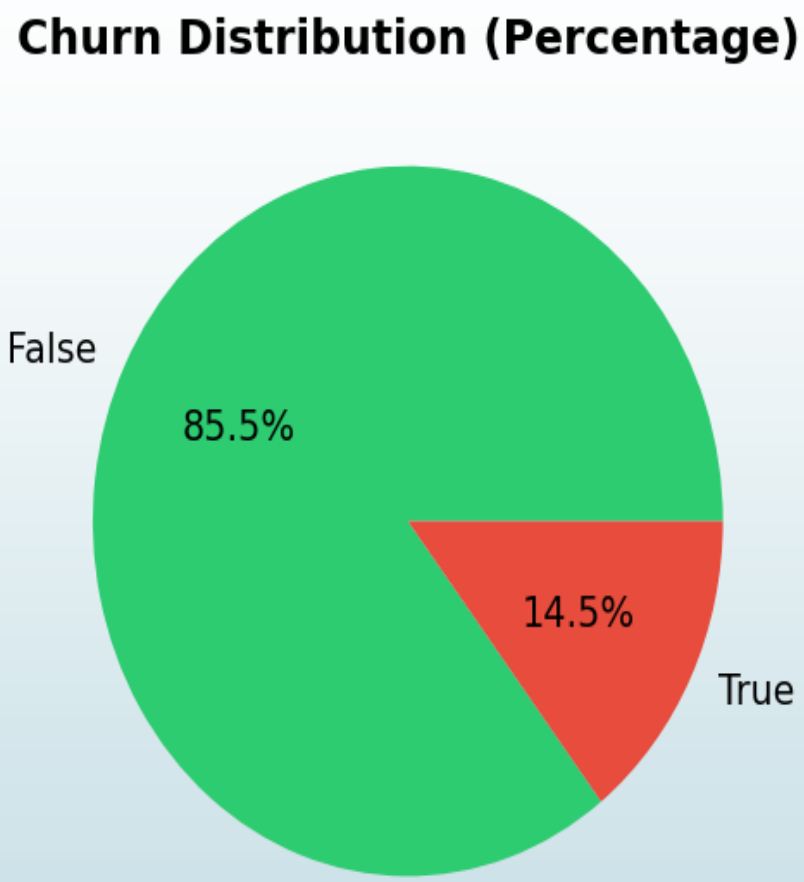
- Python 3.8+
- pandas, numpy
- scikit-learn
- imbalanced-learn (SMOTE)
- matplotlib, seaborn
- MS PowerPoint
- VS Code

Understanding data

3,333
Customers

20
Features

14.5%
Churn Rate



What the data reveals

International Plan Users

42%

churn rate

High Service Calls

4+

calls strongly predict churn

Heavy Daytime Users

Higher

bills lead to more churn

Model performance

The model catches 75 – 85% of churners

Accuracy

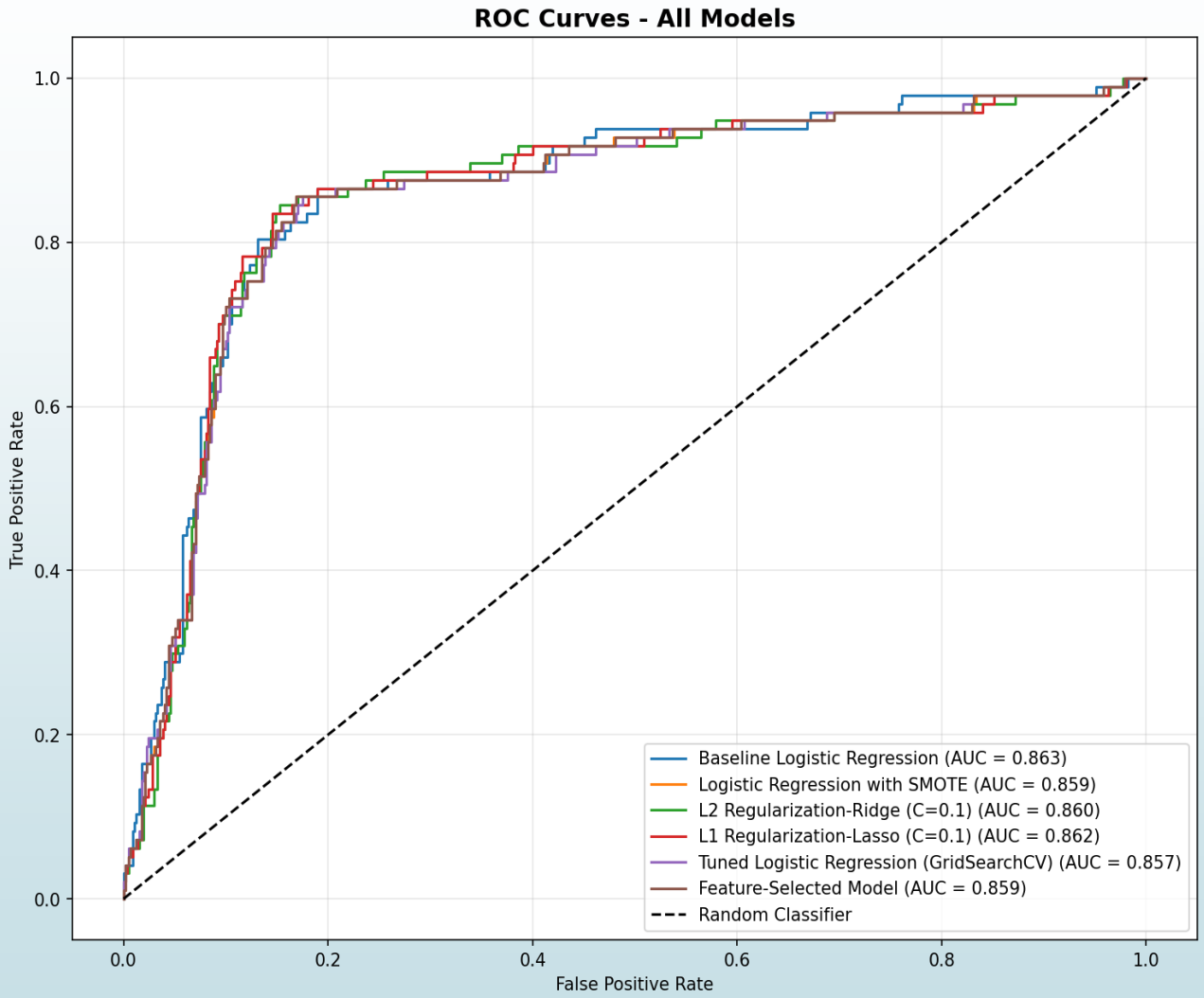
87%

Precision

55%

ROC-AUC

0.88



What drives customer churn?

- Conclusions

1	**Customer Service Calls**	*Very High*
2	**International Plan Status**	*High*
3	**Total Day Minutes**	*High*
4	**Total Day Charge**	*Medium*
5	**Voice Mail Plan**	*Medium (Protective)*

Recommendations

- **1. Immediate Actions**
- Deploy monthly churn scoring for all active customers
- Prioritize outreach to customers with 4+ recent service calls
- Offer retention incentives to high-risk segments
- **2. International Plan Review**
- Investigate why international plan customers churn at higher rates
- Consider more competitive international calling packages
- Survey international plan customers for pain points
- **3. Customer Service Quality**
- Multiple service calls are the strongest churn predictor
- Improve first-call resolution rates
- Implement automatic escalation after 3 service calls
- Monitor service quality metrics closely
- **4. Voice Mail Plan Promotion**
- Voice mail subscribers are less likely to churn
- Promote this low-cost retention tool to at-risk customers

Expected business impact:

Based on current customer base:

- 500 customers expected to churn annually
- Model identifies 400 of them (80% recall)
- 30% retention rate with intervention = 120 customers saved

Potential Annual Revenue Retention:

$120, 000

THANK YOU

QUESTIONS?